

ORIGINAL ARTICLE

Preliminary Results of Neural Networks and Zernike Polynomials for Classification of Videokeratography Maps

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ABSTRACT: *Purpose.* Our main goal in this work was to develop an artificial neural network (NN) that could classify specific types of corneal shapes using Zernike coefficients as input. Other authors have implemented successful NN systems in the past and have demonstrated their efficiency using different parameters. Our claim is that, given the increasing popularity of Zernike polynomials among the eye care community, this may be an interesting choice to add complementing value and precision to existing methods. By using a simple and well-documented corneal surface representation scheme, which relies on corneal elevation information, one can generate simple NN input parameters that are independent of curvature definition and that are also efficient. *Methods.* We have used the Matlab Neural Network Toolbox (MathWorks, Natick, MA) to implement a three-layer feed-forward NN with 15 inputs and 5 outputs. A database from an EyeSys System 2000 (EyeSys Vision, Houston, TX) videokeratograph installed at the *Escola Paulista de Medicina-São Paulo* was used. This database contained an unknown number of corneal types. From this database, two specialists selected 80 corneas that could be clearly classified into five distinct categories: (1) normal, (2) with-the-rule astigmatism, (3) against-the-rule astigmatism, (4) keratoconus, and (5) post-laser-assisted *in situ* keratomileusis. The corneal height (SAG) information of the 80 data files was fit with the first 15 Vision Science and its Applications (VSIA) standard Zernike coefficients, which were individually used to feed the 15 neurons of the input layer. The five output neurons were associated with the five typical corneal shapes. A group of 40 cases was randomly selected from the larger group of 80 corneas and used as the training set. *Results.* The NN responses were statistically analyzed in terms of sensitivity [true positive/(true positive + false negative)], specificity [true negative/(true negative + false positive)], and precision [(true positive + true negative)/total number of cases]. The mean values for these parameters were, respectively, 78.75, 97.81, and 94%. *Conclusion.* Although we have used a relatively small training and testing set, results presented here should be considered promising. They are certainly an indication of the potential of Zernike polynomials as reliable parameters, at least in the cases presented here, as input data for artificial intelligence automation of the diagnosis process of videokeratography examinations. This technique should facilitate the implementation and add value to the classification methods already available. We also discuss briefly certain special properties of Zernike polynomials that are what we think make them suitable as NN inputs for this type of application. (Optom Vis Sci 2005;82:151–158)

Key Words: videokeratography, Zernike polynomials, neural networks, artificial intelligence

The modern era of neural networks (NN's) had its beginning with the pioneering work of McCulloch and Pitts in the 1940s.¹ After a period of dormancy provoked by a skeptical view of the capabilities of NNs to solve logical problems,² the field was resuscitated by a celebrated 1982 manuscript written by Hopfield.³ The article entitled "Neural Networks and Physical Systems with Emergent Collective Computational Abilities" presented a truly revolutionary idea at the time. In this manuscript, Hopfield

was suggesting that intelligence might arise from the same laws of quantum physics that govern the atoms in a lattice structure. The idea was so revolutionary at the time "that many theoretical physicists working on spin glasses became overnight experts on the properties of neural nets."⁴

By the late 1990s, with more powerful processing units and compact low-cost mass storage devices, a more adequate scenario for better cost/benefit artificial intelligence (AI) diagnos-

tic systems finally took place. More specifically in the eye care community, at the same period, because of the growing popularity of videokeratographs as a contact lens fitting tool and also for screening patients (particularly for keratoconus) before refractive surgery, many researchers were elaborating automatic corneal diagnosis systems using different approaches from AI. Maeda et al.⁵ created a series of corneal parameters (e.g., KCI and KPI) based on different symmetries and used statistical analysis to generate algorithms for automatic detection of different corneal pathologies, including keratoconus. They showed these indices to have specificity of 97% in classifying normal from keratoconic corneas.⁶ Rabinowitz proposed the steepest radial axis (SRAX) index,⁷ which he argued to be more specific to keratoconus when used in conjunction with the central curvature (central K), superior-inferior (SI) value, and simulated keratometry (SIM K). Using these combined indices, Rabinowitz was capable of distinguishing 98% of keratoconus examinations from a group of normal control subjects. In a latter work, Rabinowitz created the KISA% index,⁸ a quantitative videokeratography algorithm for diagnosing keratoconus. Among these methods for keratoconus screening, Schwiegerling proposed the representation of corneal elevation with Zernike polynomials (ZP's)⁹ and evaluation of individual coefficients for detection¹⁰ and quantitative analysis¹¹ of keratoconus. He found that two Zernike coefficients of the low-order terms have elevated values for keratoconic corneas; he also compared this method with other indices, such as the SRAX and surface asymmetry index (SAI), and found it to perform analogously. More recently, Langenbucher et al.¹² have also used ZP's together with NNs for successful classification of keratoconus based on corneal elevation data.

In general terms what we have witnessed since the first videokeratography systems were being manufactured is a common desire of most researchers and development companies to automate processes that require much time if done manually and also to automate tasks that can usually be translated in the form of computer algorithms. The application of methods, such as statistical analysis and AI, among others, has in this sense helped the progress of algorithms for automation diagnosis of videokeratography.

In this work, such as in that of Langenbucher et al.,¹² we have also implemented an NN diagnosis method using Zernike coefficients as input parameters, but we have applied it to the classification of a broader range of corneal profiles, including keratoconus. ZP's have long been applied to the fields of optical aberration¹³ and more recently to Hartman-Shack-based wavefront systems in vision optics.¹⁴ They have also been shown to be an efficient method for fitting videokeratography corneal elevation data and also wavefront data.^{9, 11, 15}

METHODS

Representation of the Cornea As a Set of Zernike Polynomials

In videokeratography systems that use the Placido disc for measuring corneal shape and power, the examination data files are usually based on the cylindrical symmetry and number of Placido discs. In our case, an EyeSys System 2000 (EyeSys Vision, Hous-

ton, TX) was used. For this equipment, there are 16 Placido discs and 360 polar angles separated by intervals of 1°, which gives us a total of 5760 points over the cornea (corneas with excessive loss in peripheral and central discs were excluded during the random selection of the original database, and only those for which peripheral discs needed few or no editing were used in this study). (Note: The corneal topography examination process on the EyeSys software allows the user to manually edit discs that were not detected or erroneously detected at the image-processing phase. We mean by "few or no editing" those eyes that had little shadow from the eyelid and/or nose and therefore required little or no manual editing. This is an important technical aspect because excessive manual editing may change significantly important characteristics of the corneal map). Also, there are instruments on the market that have more mires than the EyeSys system, and we encourage other researchers who have different systems to conduct studies to analyze the possible differences that may arise. Nevertheless, we have chosen corneal cases with no surface singularities (i.e., extreme and abrupt changes in curvature and elevation); therefore, we do not consider that using a system with fewer Placido discs to be a considerable disadvantage compared with using a system with greater number of discs and therefore higher resolution. If dealing with cases of post-radial keratotomy or postkeratoplasty, this would certainly be a different case because these corneas have severe curvature changes; the disadvantage in these specific cases is that if one uses a dense mire, the Placido discs may overlap in certain areas of the cornea, and the image-processing algorithms could become confused in terms of the correct order in which to enumerate the processed discs (a process necessary for curvature and elevation algorithms¹⁶).

The input parameters of artificial NN are a critical aspect. This happens because each input is represented by a single neuron, and it would be an awkward task to create an NN with 5760 inputs, one for each corneal point. Our choice of input was a set composed of the first 15 Zernike coefficients, enumerated according to the Vision Science and its Applications (VSIA) Taskforce convention,¹⁷ to fit corneal elevation data files extracted from the EyeSys videokeratographer. Because many brands of videokeratography systems offer software modules for exporting elevation data or at least the documentation of how data are organized inside their files so one can build algorithms to successfully read them, any reader may perform the calculations presented here. Although the choice of using only 15 Zernike terms is usually balanced between computational power and precision, precise results for conic surfaces [root mean square error (RMSE) of 0.2 μm associated with the fitting process and 1 to 2 μm when videokeratography misalignment and improper focus are considered] have been reported without the need of more terms.^{15, 18} The quantitative benefits of using >15 terms for more complex surfaces are analyzed here in a later section. Description of the least squares fitting procedure and algorithms is not presented here because this method has been exhaustively documented elsewhere.^{9-11, 13, 15, 18}

Implementation of the NN

We have used the Matlab Neural Network Toolbox (MathWorks, Natick, MA) to implement our NN. A diagram illustrating our NN may be seen in Fig. 1.

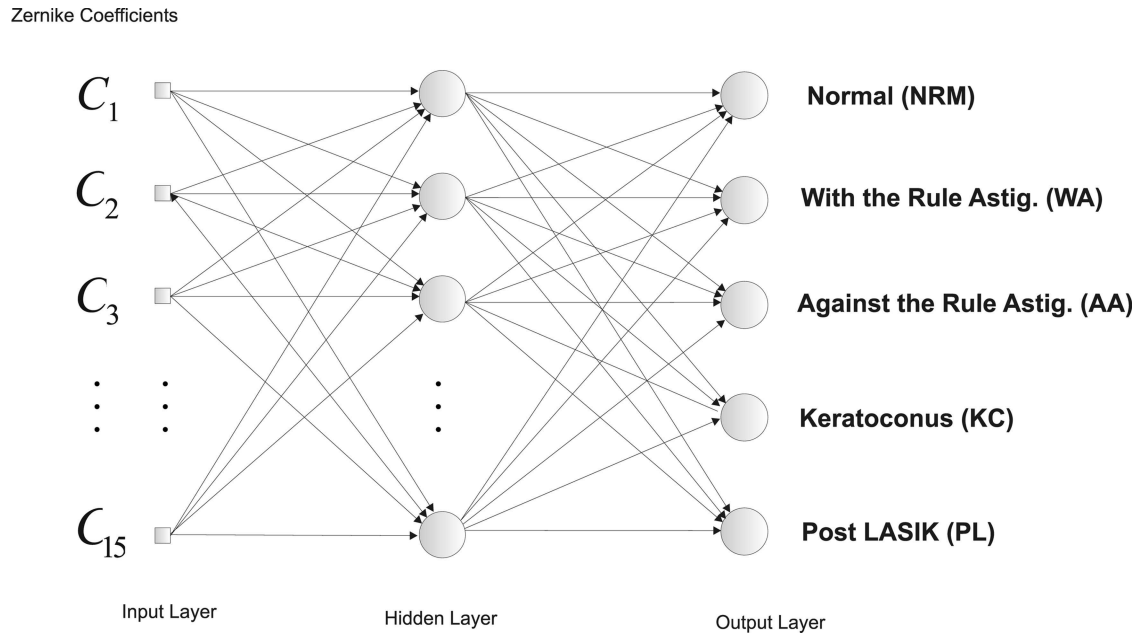


FIGURE 1.

NN implemented for classification of videokeratography examinations. As we may notice from the output layer, our NN was designed to classify five corneal shapes, which were given different abbreviations. “Normal” here refers to those corneas that have relatively small variations in curvature, similar to a spherical surface. Small astigmatisms (<0.25 D) were also considered to be “normal” corneas.

This type of NN contains all the basic elements of most NNs described in literature. It has an input layer containing neurons that receive external signals (in our case, the Zernike coefficients) and subsequent neurons in other intermediate (or hidden) layers, interacting with other layers through a series of synaptic connections. Each synapse has a given “weight,” which represents how “strong” or “weak” the connection is between any two neurons. Usually the initial stimulus travels through this complex net from the input layer onto subsequent neurons of the intermediate layers and finally reaches the output layer. This is the layer that represents the NN response to a specific input stimulus. The NN implemented here is formally defined as feed-forward with single occult layer.² After several tests using the optimization functions available in the Matlab toolbox, the intermediate layer was determined to have 10 neurons, and the output layer contains 1 neuron for each corneal shape that is being classified. There are obviously an infinite number of configurations for intermediate layers, such as the actual number of layers and the number of neurons for each layer. We tried to make the NN configuration as simple and elegant as possible for the best possible result. There is also the possibility of a much greater number of output neurons (i.e., corneal classifications), but we were restricted to the number of cases that were selected from our videokeratography database.

Such as with biological NNs, before an artificial NN can be applied to any specific problem, it must be trained first. In Fig. 2, we show a diagram illustrating the basic steps involved in the training process.

Training the NN consists of presenting it with a sequence of inputs and desired target outputs so that the synaptic weights may be adjusted to values that reflect the expected behavior. This is accomplished by comparing the target values (in our case, the training corneal maps) with the actual output values. There are

several methods for training an NN. We have applied here the “error correction” method based on the Widrow-Hoff rule,¹⁹ which consists of minimizing a cost function by adjusting the synaptic weights at each iteration. After a series of iterations (also called epochs), if the learning philosophy is the correct one, the cost function will tend to a minimum stable value, and the outputs, if the entire NN was well designed and implemented, should be close to the target values.

The testing phase consists of presenting fresh inputs to the NN after it has been trained, this time without “forcing” the output values toward target values, using the weight values of each synapse obtained during the training phase. During the testing phase, output values closer or farther from expected values will indicate success or failure of the NN. The degree of efficiency depends on several factors, such as number of intermediate layers and neurons, training philosophy, number of epochs, total time to obtain pre-established precision, and so on. There is no ideal recipe for constructing an optimal artificial NN; the NN presented here was the product of many empirical tests.

Data Acquisition

From the initial database contained in the EyeSys system, two eye care specialists selected 80 corneas of different persons according to five categories, 16 corneas for each category. In the selection process, certain criteria were used. Corneal maps had few or no nose and/or eyelid shadows; only the right eye of each patient was allowed. Right and left eyes of the same patient were not used because of the mirror symmetry of the two eyes, which could have an undesirable effect of the NN analysis; although the cases selected were not mutually exclusive (a keratoconic cornea may also manifest with-the-rule or against-the-rule astigmatism; a regular cornea

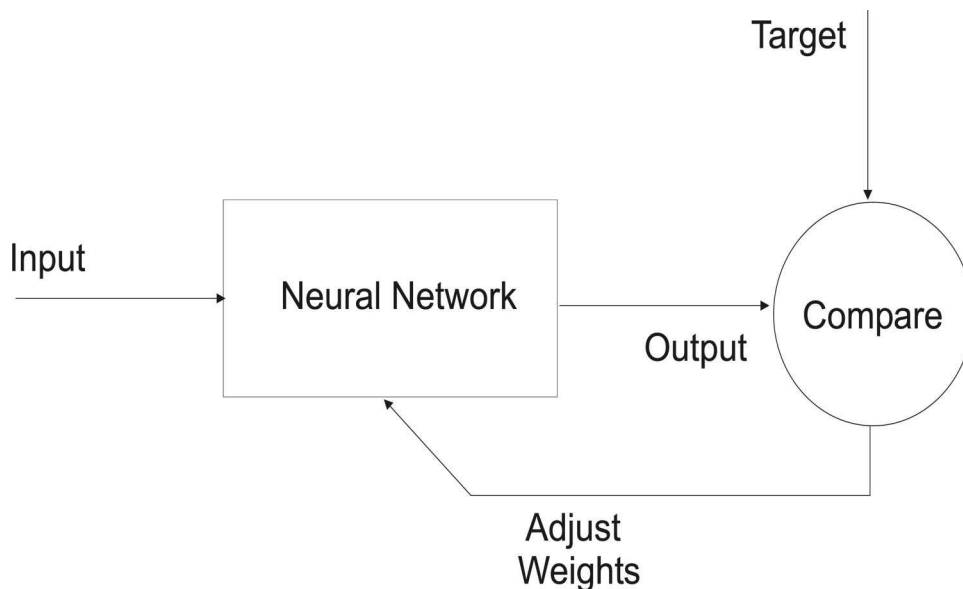
**FIGURE 2.**

Diagram illustrating the basic steps involved in training a neural network.

may also have a moderate degree of with-the-rule astigmatism, and so on), only the subjective most prevailing characteristic was used by the specialists in the classification process. Corneas such as incipient keratoconus, keratoconus with high degrees of symmetrical astigmatism, and other cases for which a single prevailing diagnosis could not be issued were excluded from the study. In the case of regular profiles, given that even the most symmetrical corneas have some degree of with-the-rule astigmatism, only corneas with SIM K < 0.25 D were considered as “regular” or “normal.” Corneal categories and respective abbreviations were (1) regular cornea (NRM), astigmatism (2) with (WA) and (3) against the rule (AA), (4) keratoconus (KC) and (5) post-laser-assisted *in situ* keratomileusis (LASIK; PL). From these, 50% (40 files) were selected as the training set, and the other 50% (40 files) were used as the testing set. This is an important step because the training phase of the NN algorithm should be accomplished by presenting equal numbers of each classification case. In this way, a more balanced solution for the synaptic weights is reached.

Each elevation data file from the videokeratography examination was transformed into a Zernike vector file containing 15 coefficient rows using the least squares method. These vectors were then fed, one by one, into our NN, and the different synaptic weights were adjusted during the iterative training in error steps of 0.01, which usually took about 300 epochs, until correct answers were obtained. Here correct answer means that a normalized value (≥ 0.5 and ≤ 1) must appear on the output neuron that represents the correct diagnosis for that specific case. When this happens, all the other neurons of the output layer must have a negative response (normalized values ≥ 0 and < 0.5); otherwise, the NN configuration is not efficient. Even when reasonable efficiency is obtained for a given NN configuration, responses for certain cases may still be dubious, such as >1 output neuron having values of > 0.5 . In such cases, the programmer must make certain assumptions and/or choices: it is obvious that a keratoconic cornea may also have astigmatism, and in this sense, the NN could be used to classify

corneas as belonging to one or more categories. In our case, we have decided that when this happens, the greater output value is the one that is used in the classification. Another possibility is to have none of the neural outputs above the 0.5 normalized threshold value. In this case, we assumed that the specific cornea did not fall into any of the classification categories and therefore was regarded as not having been successfully classified by the NN. These cases were taken into account when the statistical performance of the NN was computed afterward.

In practical terms, these are some examples of the drawbacks regarding NN implementation. Because there is no ideal recipe for building the most efficient NN, several empirical tests are required, as has been the case in the work of Accardo and Pensiero,²⁰ in terms of number of neurons in the hidden layer. The same procedure was implemented randomly to all the other examinations in the training set. When the NN obtained satisfactory results for all the eyes in the training set, their synaptic weight matrix values were saved on the computer’s hard disk. In Table 1 we may see the results of our NN for the training set.

As you may notice, our NN obtained 100% of correct responses for the training set. This showed that the NN architecture (i.e., the

TABLE 1.

Results of our NN for the 40 training corneas. 100% efficiency indicated that our NN architecture and training philosophy could also be efficient in the testing cases

Classification	NN Classification for the training set				
	NRM	WA	AA	KC	PL
NRM	8	0	0	0	0
WA	0	8	0	0	0
AA	0	0	8	0	0
KC	0	0	0	8	0
PL	0	0	0	0	8

number of occult layers and number of neurons in the occult layer), which we have varied until best results were obtained, design, and learning philosophy (feed-forward and error correction) could be in the correct direction. After the training phase, we used the other 40 files to evaluate the performance of our NN using the synaptic weights obtained during the training phase. This time the NN was fed with only the Zernike coefficients, and the output layer was expected to respond to each case. This procedure was repeated for each cornea in the testing set.

RESULTS

As mentioned in the previous section, we first used the training set to adjust the synaptic weights of the NN and then used these same weights to test our NN. The efficiency of the NN was analyzed without using any corneal data from the training set. At this point of our implementation, an output signal greater than or equal to the threshold value of 0.5 was defined as a positive response for the given category, and a value <0.5 was defined as a negative response. If more than one neuron at the output layer had response ≥ 0.5 , the one that had the greater value was considered as the correct response. If none of the responses were ≥ 0.5 , we considered that the NN did not have a classification for that corneal shape.

We analyzed the outputs for each classification category in terms of sensitivity [true positive/(true positive + false negative)], specificity [true negative/(true negative + false positive)], and precision [(true positive + true negative)/total of cases]. Table 2 shows the results for these three indices for all the test cases for each classification category. In the next section, we briefly discuss some of the basic properties of ZP's, which we think are what make them an interesting fitting method when applying AI to visual optics.

Properties of Zernike Polynomials That Make Them Attractive as NN Inputs

We believe our preliminary results, using Zernike coefficients as inputs to our NN, were positive, among other aspects, because of certain specific properties of these polynomials. These properties may be summarized by stating that ZP's form a "complete set of orthonormal polynomials" in the three-dimensional space inside the unit circle domain,^{13, 21} which guarantees that one may write any piece-wise continuous surface (which is the case for most corneas^{22, 23} that have no physical trauma, early postkeratoplasty, keratitis, or any other severe pathology that may cause abrupt changes in curvature) in space because a sufficiently large number of terms

are used. It is easy to demonstrate this fact not only for ZP's but also for other power series, such as the Fourier series. In general terms, for a limiting number of terms in a power series or a polynomial series, we should also have a limited precision in data fitting.

Fitting Precision of Zernike Polynomials

To demonstrate the variation in fitting precision as a function of the number of Zernike terms, we shall use two examples: the first is simply a spherical surface with radius of 7.8 mm, which would resemble a "normal" or "regular" cornea with no variation in curvature; the second example is a more complex surface, resembling a post-radial keratectomy (RK) cornea, for which we have used mathematical equations available in the literature.^{18, 24} Fig. 3 is a front view of computer ray-tracing simulations^{18, 24} of Placido discs for each of these surfaces.

Both surfaces were computed in Matlab, and a Zernike fit was implemented using algorithms that are extensively covered in the literature.^{13–15, 18} The RMSE is shown for both cases in Fig. 4.

As we may see in Fig. 4, the RMSE for the spherical surface converges rapidly after the third order spherical aberration term is considered (12th Zernike coefficient) in the fitting process; on the contrary, for the more complex surface, the RMSE falls less after this term and also falls less up to the 15th term. The interesting but at the same time not surprising aspect is to notice that the post-RK surface shown in Fig. 3 has an eightfold polar symmetry (i.e., there are eight wounds for an interval of 360°). Therefore, the least number of Zernike terms to minimize the fitting error would have to have at least a polar angle component of frequency of 45° . In the Zernike table, this is represented by a term that is above the 36th term listed in the VSIA convention table. The VSIA convention suggests, among other aspects, the nomenclature standers that should be used to refer to each Zernike coefficient, to avoid confusion when comparing results from different instrumentation, but does not suggest an upper limiting number of terms. Nevertheless, many researchers and manufacturers throughout the eye care community are currently using at most the first 36 terms regardless of the application. The computations shown here may suggest that the ideal number of Zernike terms should not be a fixed standard for any general surface but rather one that would allow for a minimum standard error for each specific types of surface that is being represented by ZP's.

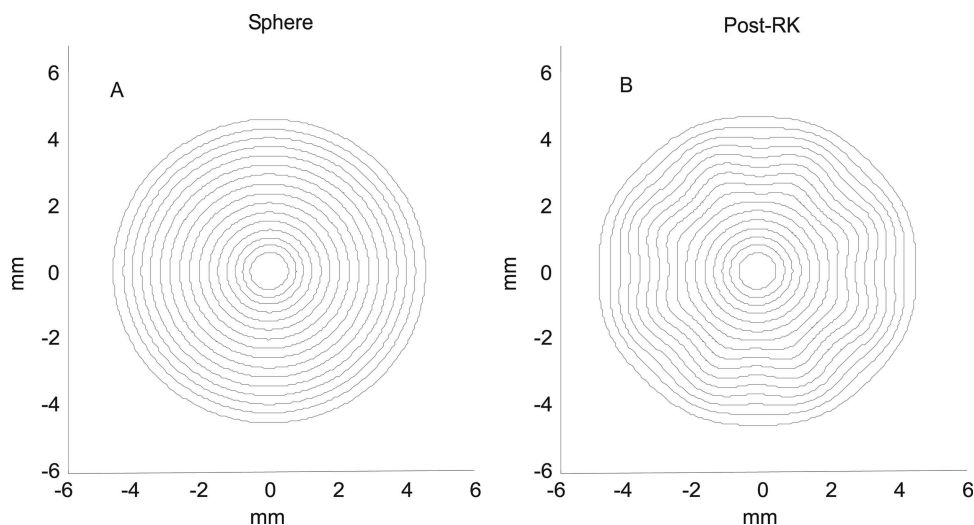
DISCUSSION

The automation of corneal topography interpretation may be an important tool for the modern practitioner, especially for those with little or no experience in the diagnosis of this type of examination. There has been, to our knowledge, two published previous attempts^{12, 25} using Zernike coefficients as inputs for NN for automation of videokeratography diagnosis. Langenbucher et al.¹² argue that Zernike decomposition of corneal topography height data allows a definition of a robust algorithm for detection of keratoconus; Smolek et al.²⁵ argue that their attempts resulted in a poor diagnosis tool when Zernike coefficients were used alone.

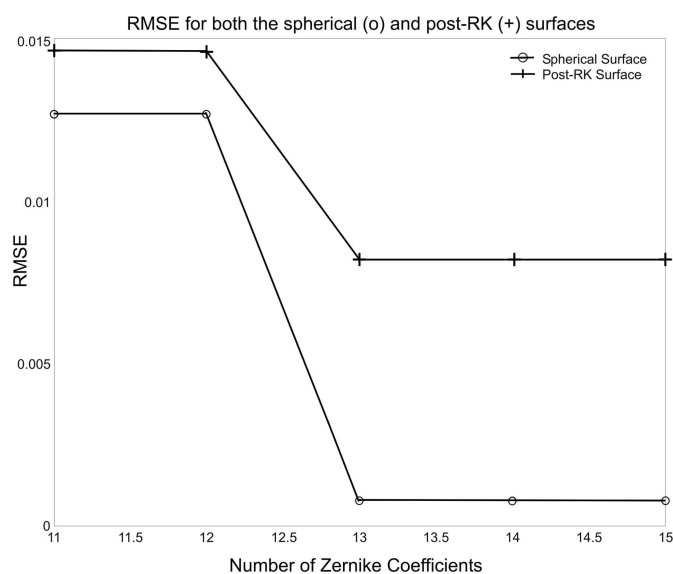
A possible answer to unsuccessful results, among a series of other technical differences that one or other method may have in terms

TABLE 2.
Results for the 40 test corneas

Classification	Sensitivity (%)	Specificity (%)	Precision (%)
NRM	56.25	95.31	87.50
WA	56.25	98.44	90.00
AA	87.50	96.88	95.00
KC	93.75	98.44	97.50
PL	100.00	100.00	100.00
Mean	78.75	97.81	94.00

**FIGURE 3.**

Synthetic images of Placido mires for a (A) spherical surface and a (B) post-RK surface. Algorithms for generating such surfaces and their Placido image are shown elsewhere.^{18, 24}

**FIGURE 4.**

RMSE as a function of the number of Zernike terms for both surfaces shown in Fig. 3. Because differences between both surfaces are subtle up to the third-order polynomials, we have plotted only the RMSE from the 11th to 15th coefficient, which are the fourth- and fifth-order terms. Again, it may be noticed that the more symmetrical surface's RMSE tends to zero more rapidly than the post-RK, as expected. Moreover, we may also notice that for only 15 Zernike terms, precision for the post-RK surface does not go to $<5 \mu\text{m}$, clearly showing that for such a surface this is not the ideal number of terms according to acceptable videokeratography sag precisions available in the literature.^{18, 24} Observation: the x axis above shows fractional numbers that are obviously impossible; this only happens because of the Matlab graph plotting style.

of the NN implementation details and complexities of corneal surfaces used for training and testing, may be the number of Zernike coefficients used in data fitting. As we have shown here, the number of terms is extremely relevant to the complexity of the surface, and using a limited number of them for complex surfaces may “hide” visually significant corneal aberrations.²⁶ Although not

conclusive because of the low number of corneas (80) and types of corneas (5), in this work we have found that the first 15 Zernike coefficients are sufficient for obtaining satisfactory results at least for simple and symmetric corneal anomalies. This is in agreement with Schwiererling's findings that Zernike coefficients may be used to describe corneal surface anomalies.^{9–11}

The mathematical arguments given in the previous sections allow us to understand why the number of Zernike terms is important. We have used simple corneal symmetries here in our NN; therefore, the use of 15 terms was sufficient. If one wants to train and test the NN with more complicated corneal surfaces (more complex symmetries), then higher-order polynomials should be used, or a mixture of different methods, given that a high number of Zernike is computationally intensive. In our database, we had no cases of post-RK, but if one wants to build an NN that also interprets this corneal shape, the number of inputs should be sufficiently high (as we have shown in Fig. 4; for errors $<5 \mu\text{m}$, >5 Zernike terms are needed such that the Zernike angular frequency can represent the post-RK wound scars).

With the help of automated classifiers, used separately or blended together, the eye care professional, with the accumulation of different cases, could acquire more experience in less time; and for those more experienced users, the machine interpretation would be helpful as a “second opinion.” Other undesirable aspects that influence human judgment capabilities, such as the emotional factor, concentration, memory, and so on, are not present in the artificial system, which makes it less subjective.

Automatic diagnostic systems may be built using other methodologies, such as variational analysis, decision trees, and genetic algorithms. Each method has advantages and disadvantages of its own. The NN approach is usually simple to implement in terms of algorithms but is computationally expensive during the training process. It also has a poor handling of missing data and generalization beyond training experience, which may happen when it is presented with a case that it has not been trained to classify. Nevertheless, NNs are powerful for detecting hidden characteristics of the input data, even if those are not formulated as specific input.

NNs are usually based only on the input data, such that no extra information is necessary regarding rules, interrelation of parts that compose the problem, or any other logical structure. The use of different approaches is perhaps the most solid base to generate reliable software for automation of complex decisions and classifications.

To design an efficient NN, one of the critical steps is the choice of what and how the input data will be fed. In this work, we have chosen to use a set of 15 Zernike coefficients, but this does not mean that they are sufficient as inputs for any type of cornea. Swiegerling has shown the efficiency of Zernike coefficients as descriptors of specific corneal anomalies such as keratoconus,^{10, 11} and Guirao et al.¹⁵ demonstrated its precision in fitting synthetic conical surfaces (RMSE errors of 0.2 μm using 15 coefficients). Nevertheless, for surfaces of more complex symmetries such as post-RK, the ideal number of Zernike terms should be >15 or even 36 terms, if one wants to obtain the same order of precision as those seen in conic and spherocylinder surfaces.¹⁸ This may impose a limiting factor to the use of Zernike coefficients for complex symmetry surfaces because computational time quadruplicates every time the number of coefficients double.

Our NN was trained using 40 corneas with five distinct characteristics and was then tested on another group of 40 corneas not used in the training process. The mean value of precision was 94%. Although we have used only 80 corneas in this study, we consider these results promising. Nevertheless, a greater number of corneas and corneal pathologies should be considered and also other NN configurations, such as a variation in the number of neurons in the hidden layer, and different training philosophies. NNs applied in other fields of medicine, such as ultrasound image interpretation,²⁷ cancer diagnosis,²⁸ and visual field interpretation,²⁹ have shown efficiencies similar to those shown in this work. Unacceptable errors, such as a keratoconic cornea being interpreted as a “normal” cornea, have not occurred.

Certain improvements in the NN architecture and training are possible. The dependency of the synaptic weights with the number of iterations for NN training suggests that a greater database will provide more reliable results regarding precision, specificity, and sensitivity. Also, other authors have shown a close relation between the number of hidden layers and the efficiency of NNs.²⁰

Our claim is that an AI system for corneal diagnosis that uses Zernike terms as input for an NN should also be considered in future automation systems. First, it is easy to implement, given that ZP's and least squares fitting methods are well documented in literature and have been used for decades in the field of optical engineering.¹³ Second, it also allows, as do other methods,^{30–32} any interested reader to reproduce these results in the laboratory. The proposed method also represents an elegant mathematical approach in terms of corneal representation and may serve well as input into a well-established NN theory and algorithms.

We also believe that a commercial version of this system, if implemented, could help the practicing eye care professional in better understanding the relation between certain corneal characteristics and the different Zernike terms. Something awkward these days is trying to associate each Zernike coefficient value with the corneal characteristic just by visual inspection.

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